Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Re: Memory Rank Paper - My Response to Your Latest Comments

Ezequiel Boero <ezequiel.boero@adagioinstitute.org> Mon, Sep 28, 2026 at 9:56 PM
To: "Benjamin D. Summers" <bsummers@adagiollc.com>
Cc: "ISMAIL-SUTTON, SARA R." <sara.r.ismail-sutton@durham.ac.uk>

Ben,

This also closes the question and comment I have on the memory rank paper.
I verified each of your factual statements against revision 2.1, and all of them hold: the algebraic-decomposition clause is gone;
Open Problem (v) is unchanged; the self-authored references are [27]–[30] (instant, Koopman, convex perturbation, selection functionals)
with the memory production paper absent; and the class exchange rate survives as the closed form of Example 8.4.

Two things follow and they correct my own record.

Theorem 5.11. Your restatement is right: the open converse is that of **(a)**, from $\kappa_J(V) = 0$ to $V \in \mathcal{F}_{D_{\max}}$,
and **(d)** is precisely what closes the complementary case when some optimal law has positive one-time marginals.
My record had labelled the gap "of Theorem 5.11(d)", which inverts the two parts.
The statement of the limit was correct and only the label was wrong; it is corrected.

Open Problem (v), and a correction in your favour. You withdrew the clause because no result of the paper proves it.
I went further and checked what the companion actually establishes, and the position seems to be stronger than you argued.
Corollary 4.8 of the memory production paper gives two-sided bounds at the charge level and then states, in its own words,
"these are bounds; the charge does not satisfy an additive identity of its own."
Since Open Problem (v) asks for a decomposition of the metric charge $h_{n'}(\Lambda V)$, and the companion expressly declines
the additive identity at exactly that level (with a balance law for the full profile left open as its own Open Problem (ii))
Open Problem (v) is correctly posed as open. My record had said it "still poses as open" something "which is exactly the balance law"
of the other manuscript. That was wrong, and I have rectified it.

What survives is narrower, and I want to state it precisely because it is the only substantive item I am carrying forward.
Open Problem (v) describes the deficit level by saying that *at that level the decomposition is a question about rational functions*
of the fine kernels and marginals; and at that level the companion proves it outright: Theorem 4.2(i) gives the exact matrix identity
$\Delta V^\lambda = \Phi_\lambda + P_\lambda$, Theorem 4.2(ii) gives the contractivity $\|\Phi_\lambda\| \leq \|\Delta V\|$
by orthogonal splitting, and Theorem 4.5 characterises the zero set of P_λ.

Limits. You restate three of the five. The other two stand unchanged and unobjected: the empirical content, now vacuously compliant
since the manuscript contains no computation; and the two results not adopted with reasons given.

I consider the report of the paper closed with no outstanding amendments nor further comments.
No mathematical deficiency remains so far I can notice.

Ezequiel

[Quoted text hidden]